Consent Form

Title of research study: *Expert Evaluation of Counterfactual Explanations for Postprandial Blood Glucose Management*

Primary Investigator: Dr. Stephanie Marita Carpenter, Assistant Professor in the College of Health Solutions at Arizona State University

Co-Investigator: Dr. Hassan Ghasemzadeh, Professor in the College of Health Solutions at Arizona State University

I am Abdullah Al Mamun, a graduate student under the direction of Dr. Hassan Ghasemzadeh and Dr. Stephanie Marita Carpenter in the School of Computing and Augmented Intelligence and College of Health Solutions at Arizona State University. I am conducting a research study to evaluate MealCF, an explainable artificial intelligence (AI) system that predicts whether a person's post-meal (postprandial) blood glucose will reach 140 mg/dL or higher and generates counterfactual explanations - brief recommendations describing small dietary or behavioral changes that could lower the postprandial blood glucose level.

You are considered invited to take part in the survey if one of the research team members reached out to you and invited you in writing to take part in the main survey. Your participation will involve completing a one-time online survey that takes approximately 30 minutes. After a few brief questions about your area of expertise, you will review 8 de-identified cases. Each case describes a person whose postprandial blood glucose reached 140 mg/dL or higher, along with a MealCF recommendation intended to help avoid that outcome. For each recommendation, you will answer rating questions about its effectiveness, feasibility, actionability, and acceptability, and you may add optional written comments. The cases are de-identified research scenarios, not your patients or clients; you are not being asked to provide care, diagnose anyone, make treatment decisions, or follow any recommendation yourself, and MealCF recommendations should not be treated as medical advice. You can withdraw consent and stop participation at any time before submitting the survey.

Your participation in this study is voluntary. You must be 18 or older to participate. If you choose not to participate or to withdraw from the study at any time, there will be no penalty. If you complete the survey, we will send you an Amazon gift card with a value of $25. After our research team verifies the receipt of your survey response, the gift card will be sent to your email address that was used to communicate with us.

There is no direct benefit to you from participating, although your feedback may help improve health-related AI systems and the way their recommendations are presented. There are no foreseeable risks or discomforts to your participation.

Your responses will be confidential. Your contact information will be used only for compensation and administrative tracking. Your email address and any other identifying information will be kept confidential and no one other than designated research team members will be able to track the main survey response to you. The results of this study may be used in reports, presentations, or publications but your name or contact information will not be used. De-identified responses collected in this survey may be shared in publications. Additionally, data from participants will be de-identified and used for possible future investigations. De-identified responses may be used by AI models for processing or summarizing and they may be used to improve AI models.

If you have any questions concerning the research study, please contact graduate student Abdullah Al Mamun at a.mamun@asu.edu (425-615-4157), the Primary Investigator Dr. Stephanie Marita

Carpenter at Stephanie.M.Carpenter@asu.edu (480-727-5740) or Co-Investigator Dr. Hassan Ghasemzadeh at Hassan.Ghasemzadeh@asu.edu (480-884-2639). If you have any questions about your rights as a subject/participant in this research, or if you feel you have been placed at risk, you can contact the Chair of the Human Subjects Institutional Review Board, through the ASU Office of Research Integrity and Assurance, at (480) 965-6788. Please let me know if you wish to be part of the study by checking the consent form box in the main survey.